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## Ex.No.6: 5G NR Frame Structure Analysis with Different Subcarrier Spacing

### Objective 1:

To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.

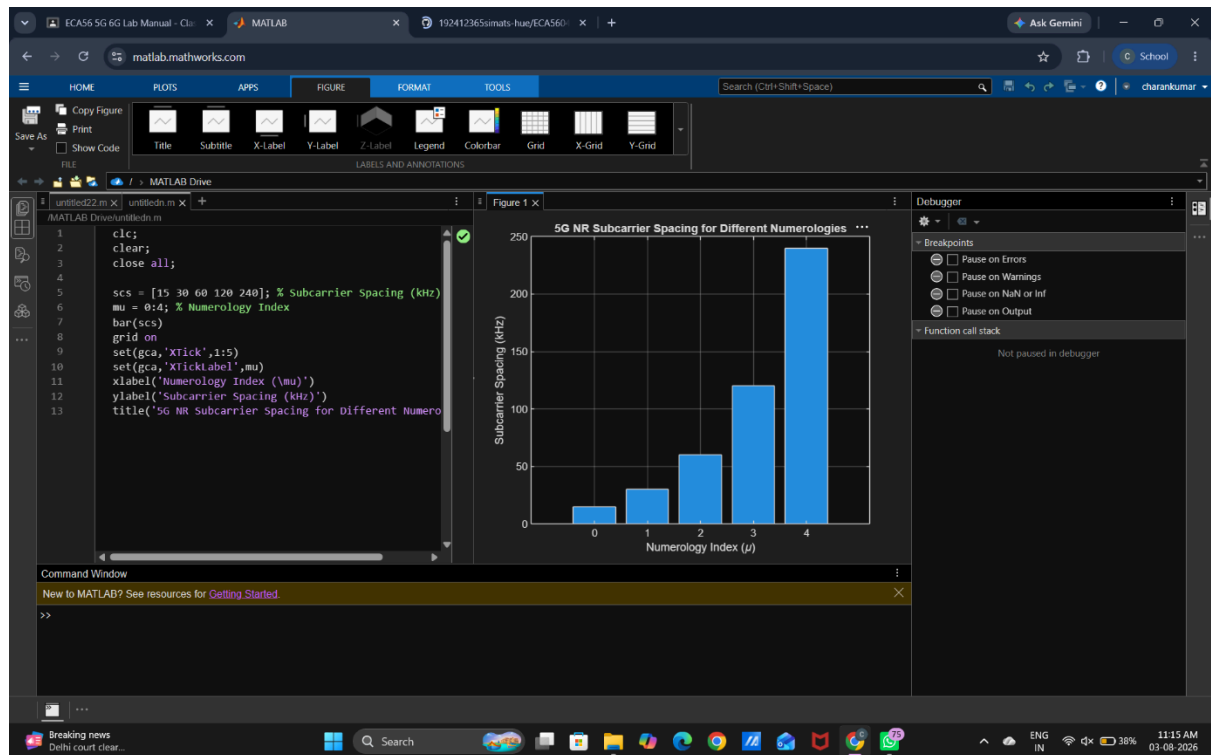
### Program (Matlab Code)

```
clc;
clear;
close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```



## Output:



## Objective 2:

To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.

## Matlab Code

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
```

```
plot(scs, slot, 'o-', 'LineWidth', 2)
```

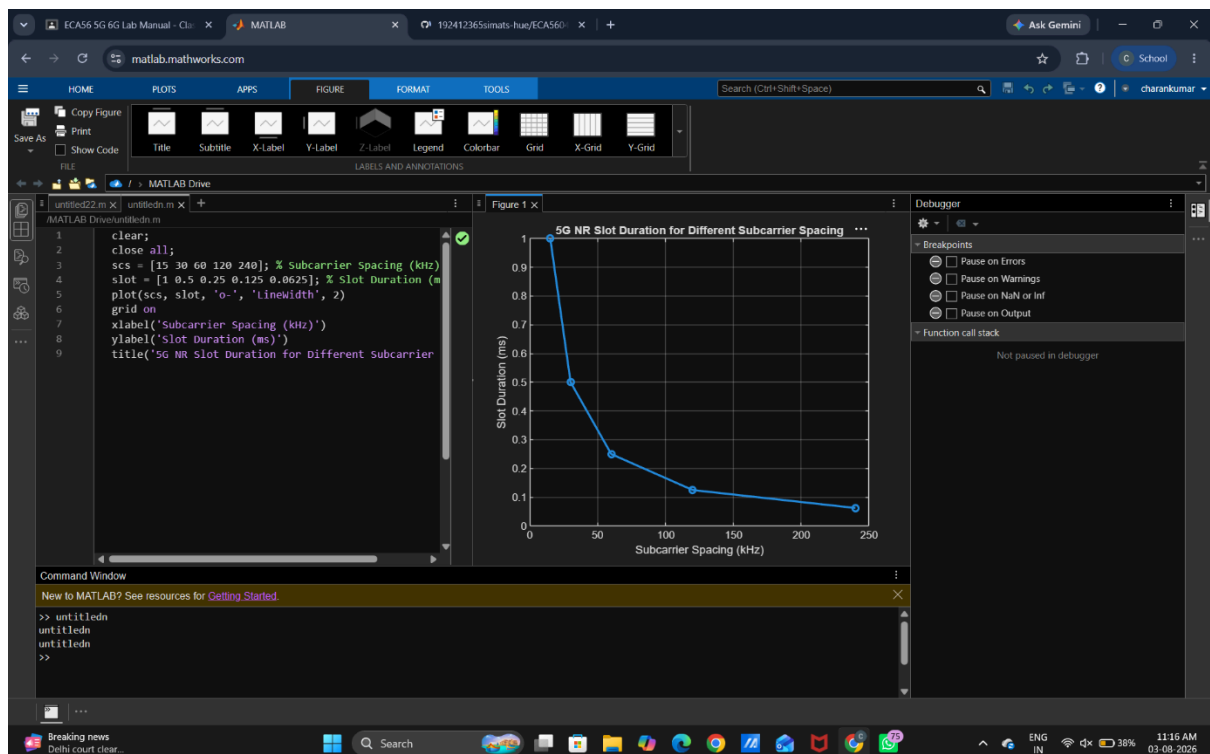
```
grid on
```

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Slot Duration (ms)')
```

```
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

**Output**



### Objective 3:

To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

### Matlab Code

```
clc;
```

```
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```



# Output

